

How to deal with regression to the mean in intervention studies

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A randomised trial published in 1995 compared, in adults with serum cholesterol concentrations of 5–8 mmol/L, the efficacy of three types of diet and lifestyle intervention in lowering plasma lipid levels.¹ After 6 months, the cholesterol concentration had fallen by around 10% in all three groups. This improvement was attributed to the interventions, but since some reduction in cholesterol would have been expected due to regression to the mean (RTM), the true efficacy of the interventions must remain unknown. So, what is regression to the mean and why does it happen?

RTM is, in essence, a chance finding masquerading as a real one. Despite the name, it is not confined to regression analysis but turns up in several different situations.² This paper is concerned with intervention studies. Such studies often target individuals who have unusually high values of a risk variable, such as cholesterol. In a group of such individuals, the mean cholesterol level will, on remeasurement, be lower than the starting mean, even without any intervention or treatment.^{3–6} This is the RTM effect. In a group selected on the basis of unusually low values of a variable RTM works in the other direction.

RTM occurs with any variable that fluctuates within an individual, either genuinely or due to measurement error.⁷ Examples are cholesterol, blood pressure, and plasma glucose. A single measurement may be higher or lower than the individual's long-term average (or "true") value, but individuals selected because they have a single high measurement will include a disproportionate number whose measurement was by chance higher than their true value. This can be better understood from the figure. **The population distribution of the measurement in question consists of numerous individual distributions, with the peak of each representing the individual's true value.** Above any high cut-off point (the vertical line at L) there will be more individuals like A (with their long-term average below the line) than like B (with their long-term average above it). On remeasurement, the values recorded for these individuals will tend to revert towards their long-term average, so that the mean measurement of the group as a whole will fall.

In this paper we illustrate the magnitude of the RTM effect in a group of individuals with high cholesterol concentrations, selected from a population with repeated cholesterol measurements and no intervention. We then describe two methods for estimating the RTM effect in an intervention study, and go on to show how intervention

studies can be designed specifically to limit the RTM problem.

How large is RTM effect?

RTM observed with no intervention

We observed the RTM effect in repeated measurements of total serum cholesterol on patients in the UK Prospective Diabetes Study (UKPDS).⁸ Measurements were done at five annual visits for 1914 men aged 25–65 (mean 52). There was no trend in mean cholesterol over this period (overall mean 5.25 mmol/L; range of annual means 5.21–5.29 mmol/L).

For the purpose of this analysis a "high-risk" group was defined to include those whose cholesterol at visit 1 was 6.50 mmol/L or more (n=189). The mean cholesterol concentration in this group was 7.21 mmol/L but at visit 2, 12 months later, the mean in the same men was 6.55 mmol/L. This 9% fall (0.66 mmol/L) is due to RTM.

RTM effect estimated by formula

In an intervention study the RTM effect cannot be observed because it is mixed with any genuine intervention effect. It can, however, be estimated. The following method was described more than twenty years ago.^{3,4,9}

It is assumed that everyone has a true cholesterol around which his or her measured value varies randomly with the same degree of variability (figure). This variability means that there is a less than perfect correlation between two measurements of cholesterol made on the same individual at different times. **The greater the within-person variability the weaker the correlation.** Both the individual and population distributions are assumed to be normal.

Estimation of the RTM effect requires knowledge of the population mean and SD. This can be assumed from

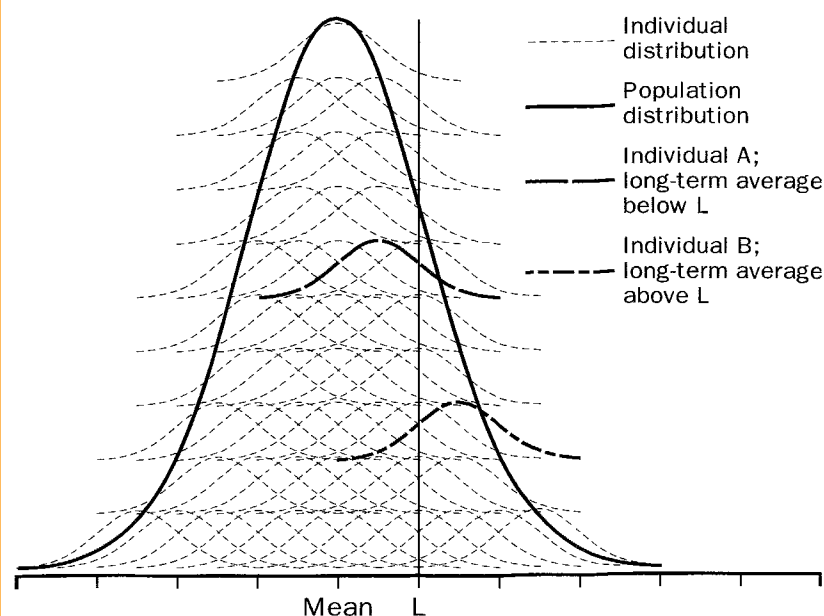


Figure: Schematic distributions of serum cholesterol in populations and in individuals

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	Serum cholesterol (mmol/L)
Values assumed	
Population mean	5.21
Population variance	1.03
Population SD	1.02
Correlation between repeated measurements	0.67
Variance of distribution of true values	0.69
Within-person variance	0.34
Results observed	
Observed mean on 1st measurement	7.21
Observed mean on remeasurement	6.55
Observed RTM	0.66
Results based on formula	
Expected mean on 1st measurement	6.98
Expected mean on remeasurement	6.40
Expected RTM	0.58
Results in simulation	
Estimated mean on 1st measurement	6.97
Estimated mean on remeasurement	6.41
Estimated RTM	0.56

Table 1: RTM effect observed, estimated by formula, and estimated by simulation, for group selected on basis of first cholesterol measurement ≥ 6.50 mmol/L

previous work or estimated from a single screening measurement in the population. We used the mean (5.21 mmol/L) and SD (1.02 mmol/L) derived from the first visit in the UKPDS.

The correlation between repeated measurements in an individual is also required. This will depend on the nature of the within-person variation—is it predominantly biological or due to measurement error?—and, possibly, on the interval between measurements.^{10,11} For an intervention study the relevant interval is that between the initial measurement and the remeasurement after intervention. In the UKPDS, for measurements taken a year apart, correlation coefficients were about 0.70. Higher values have been reported from other studies.^{10,12} The correlation cannot be estimated in a straightforward way from the high-risk group alone¹³ because restricting the values of the correlated variables will bias the correlation coefficient downwards.¹⁴

The calculation of the RTM effect is then done as follows:

Cholesterol is assumed to be normally distributed, with mean μ and SD σ . Let the cut-off point for selection be k .

Let $z=(k-\mu)/\sigma$. . . from the values in table 1, $z=1.26$ ($[(6.50-5.21)/1.02]$)

Let $c=(\text{ordinate of normal distribution at } z)/(\text{area under normal curve beyond } z)$. . . when $z=1.26$, $c=0.1804/0.1038=1.74$ (see table 2).

Expected mean cholesterol for people with first measurement greater than k is $\mu+c\sigma$. . . here, this is 6.98 mmol/L ($5.21+[1.74 \times 1.02]$).

Expected mean on remeasurement is $\mu+\rho c\sigma$, where ρ is the correlation between repeated measurements in an individual . . . based on results for visits 1 and 2, $\rho=0.67$ so expected mean is 6.40 mmol/L ($5.21+[0.67] \times [1.74] \times [1.02]$).

Expected RTM effect is $c\sigma(1-\rho)$. . . here, this is 0.58 mmol/L ($6.98-6.40$).

So, the more extreme the cut-off point and the weaker the correlation between repeated measurements the greater the effect of RTM.

The expected mean value is slightly lower than the observed mean (table 1) possibly because a normal distribution is assumed whereas the actual distribution was skewed to the right. The estimated and observed RTM effects were 0.58 mmol/L (8%) and 0.66 mmol/L (9%), respectively.

RTM effect estimated by simulation

Simulation is based on the same assumptions as the formula (above) and requires the same information but expressed differently. You will need a computer package that can generate observations from a normal distribution with known mean and variance. Two sets of observations are generated, representing first and second measurements in the population. Cases with a first measurement above the cut-off point are selected and the mean first and second measurements in this group are calculated. The difference is the RTM effect. The method is as follows:

Population variance σ^2 (1.03 [mmol/L]² in our example, table 1) has two components: the variance of “true” cholesterol measurements (δ^2) and the variance of repeated measurements in an individual (γ^2). $\delta^2=\rho\sigma^2(0.67 \times 1.03=0.69)$; and $\gamma^2=\sigma^2-\delta^2$ ($1.03-0.69=0.34$).

Using the SAS package¹⁵ 1914 values, representing a set of true cholesterol measurements, were generated from a normal distribution with mean 5.21 (μ) and variance 0.69 (δ^2). Values representing a set of actual first measurements were generated by adding to each true value a random error from a normal distribution with mean zero and variance 0.34 (γ^2). A set of actual second measurements was generated by adding a different random error to each true value. Cases with a first measured value of 6.50 mmol/L or above were selected, and the mean first and second measurements in this group calculated. The difference between these two means estimates the RTM effect.

If the same assumptions and values are used simulation should produce the same results as the formula method, except for sampling variation. However, simulation is more flexible. For example, a nutrition education programme in Bergen, Norway, was open to the general public without regard to cholesterol level, but individuals with a high cholesterol level were more likely than others to return for remeasurement.¹⁶ The formula method could not be used in this situation, but the RTM effect was estimated by simulation, the probability of remeasurement being related to initial cholesterol level.¹⁷

Simulation on our data produced similar results to those obtained with the formula method (table 1).

Designing studies to minimise RTM problem

We suggest three approaches:

Randomised controlled trials

If trial participants are randomised to active or control treatment any difference in response between the two groups will exclude RTM. However, the response in both

z	0.00	0.02	0.04	0.06	0.08	0.10	0.12	0.14	0.16	0.18
0.00	0.80	0.81	0.82	0.84	0.85	0.86	0.88	0.89	0.90	0.92
0.20	0.93	0.94	0.96	0.97	0.98	1.00	1.01	1.03	1.04	1.05
0.40	1.07	1.08	1.10	1.11	1.13	1.14	1.16	1.17	1.19	1.20
0.60	1.22	1.23	1.25	1.26	1.28	1.29	1.31	1.32	1.34	1.35
0.80	1.37	1.38	1.40	1.41	1.43	1.45	1.46	1.48	1.49	1.51
1.00	1.53	1.54	1.56	1.57	1.59	1.61	1.62	1.64	1.65	1.67
1.20	1.69	1.70	1.72	1.74	1.75	1.77	1.79	1.80	1.82	1.84
1.40	1.85	1.87	1.89	1.90	1.92	1.94	1.96	1.97	1.99	2.01
1.60	2.02	2.04	2.06	2.08	2.09	2.11	2.13	2.15	2.16	2.18
1.80	2.20	2.21	2.23	2.25	2.27	2.28	2.30	2.32	2.34	2.36
2.00	2.37	2.39	2.41	2.43	2.44	2.46	2.48	2.50	2.52	2.53
2.20	2.55	2.57	2.59	2.61	2.62	2.64	2.66	2.68	2.70	2.71
2.40	2.73	2.75	2.77	2.79	2.80	2.82	2.84	2.86	2.88	2.90
2.60	2.91	2.93	2.95	2.97	2.99	3.01	3.02	3.04	3.06	3.08
2.80	3.10	3.12	3.13	3.15	3.17	3.19	3.21	3.23	3.25	3.26
3.00	3.28	3.30	3.32	3.34	3.36	3.38	3.39	3.41	3.43	3.45
3.20	3.47	3.49	3.51	3.53	3.54	3.56	3.58	3.60	3.62	3.64
3.40	3.66	3.68	3.69	3.71	3.73	3.75	3.77	3.79	3.81	3.83
3.60	3.85	3.86	3.88	3.90	3.92	3.94	3.96	3.98	4.00	4.02
3.80	4.04	4.05	4.07	4.09	4.11	4.13	4.15	4.17	4.19	4.21
4.00	4.23									

Table 2: Values for c for different values of z

groups will be inflated by RTM so the actual size of the treatment and control effects will remain unknown.¹

Basing selection on several measurements

Since RTM is more pronounced when the variability in repeated measurements is greater, RTM can be reduced by using the mean of several measurements, rather than a single one, as the basis for selection. Some intervention studies have used this approach.¹⁸⁻²⁰

Using UKPDS data we examined the reduction in the RTM effect as selection was based on the mean of two, three, or four measurements (visits 1 and 2, visits 1, 2, and 3, and visits 1, 2, 3, and 4) with remeasurement the following year. With selection based on the mean of four measurements instead of a single measurement, the observed RTM effect was reduced from 0.66 mmol/L (9%) to 0.21 mmol/L (3%). Little benefit (and much inconvenience) accrues from basing selection on more than four measurements.⁹

The method of estimating the RTM effect for multiple preintervention measurements^{3,9} is as follows:

Let $\bar{y}$ be the mean of n measurements ($\bar{y}$ is normally distributed with mean μ and variance $\delta^2 + \gamma^2/n$).

Let $z' = (k - \mu) / \sqrt{(\delta^2 + \gamma^2/n)}$. . . for $n=4$, $z'=1.47$ $[(6.50 - 5.21) / (0.69 + 0.34/4)]$.

Expected mean cholesterol for those with the mean of n measurements greater than k is $\mu + c' \sqrt{(\delta^2 + \gamma^2/n)}$. . . mean is thus 6.89 mmol/L $(5.21 + [1.91] \times [0.88])$.

Let $c' = (\text{ordinate of normal distribution at } z') / (\text{area under normal curve beyond } z')$. . . when $z'=1.47$ $c'=0.1354 / 0.0708 = 1.91$ (see table 2).

Mean cholesterol on remeasurement is $\mu + c' \delta^2 / \sqrt{(\delta^2 + \gamma^2/n)}$. . . this is 6.71 mmol/L $(5.21 + [1.91] \times [0.69] / 0.88)$.

Expected RTM effect is the difference between the two means . . . here 0.18 mmol/L $(6.89 - 6.71)$.

The estimate for four measurements (0.18 mmol/L) is close to the observed value (0.21 mmol/L).

Selecting on one measurement and assessing treatment effect from another

In 1972 Ederer¹² suggested that RTM can be avoided if two measurements are taken before the intervention. The first is used as the basis for selection, the second as the baseline from which to assess the intervention effect. Since RTM has already taken place, the argument goes, the mean improvement measured from this baseline is the true effect of the intervention. This approach was used in the Minnesota Coronary Dietary Survey Trial to assess the relation between starting level of cholesterol and response to treatment. The trial showed that even after RTM had been taken into account, those with the highest cholesterol levels were the most responsive to dietary intervention.²¹

This simple approach is attractive but it eliminates RTM entirely only if the correlation between first and second measurements is the same as that between first and third.^{9,22} In the UKPDS data, mean cholesterol levels

at the first, second, and third measurements for the group with a first measurement of 6.50 mmol/L or above were 7.21 mmol/L, 6.55 mmol/L, and 6.63 mmol/L. RTM was therefore eliminated between the second and third measurements, a result consistent with the fact that in our data, the correlation between first and second measurements (0.67) was the same as that between first and third measurements (0.68).

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